Licenciatura em Engenharia Informática
ALGAN
1st Semester 2025-2026
TP2

1. Consider the following matrices

$$\mathbf{A}=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \quad \mathbf{B}=\begin{bmatrix}0 & 2 & 0\\ 0 & 0 & 3\\ 0 & 0 & 0\end{bmatrix} \quad \mathbf{C}=\begin{bmatrix}1 & 1 & 2\\ 1 & 0 & 1\\ 0 & 1 & 3\end{bmatrix}$$
$$\mathbf{D}=\begin{bmatrix}-1 & 0 & 1\\ 0 & 1 & 2\\ 1 & 0 & 1\end{bmatrix} \quad \mathbf{E}=\begin{bmatrix}a & b & c\\ b & a & b\\ c & b & a\end{bmatrix} \quad \mathbf{F}=\begin{bmatrix}1 & 1 & 1\\ x & y & z\\ x^2 & y^2 & z^2\end{bmatrix}.$$

Compute:

a) $\det(\mathbf{A})$; **b**) $\det(\mathbf{B})$; **c**) $\det(\mathbf{B}-\lambda\mathbf{I}_3)$; **d**) $\det(\mathbf{C})$; **e**) $\det(\mathbf{D})$;

f) $\det(\mathbf{C}\,\mathbf{D})$ **g**) $\det(\mathbf{C})\det(\mathbf{D})$ **h**) $\det(\mathbf{D}\,\mathbf{C})$ **i**) $\det(\mathbf{C}^T)$ **j**) $\det(\mathbf{B}-\mathbf{C})$;

k) $\det(\mathbf{C}^2)$ **l**) $\det(\lambda\,\mathbf{A})$ **m**) $\det(\lambda\,\mathbf{D})$ **n**) $\det((\mathbf{C}\,\mathbf{D})^4)$ **o**) $\det(\mathbf{C}+\mathbf{D})$

p) $\det((\mathbf{C}\,\mathbf{D})^{-3})$ **q**) $\det(\mathbf{C}^{-1})+\det(\mathbf{D}^{-1})$ **r**) $\det(\mathbf{E})$ **s**) $\det(\mathbf{F})$

2. Consider the matrix $\mathbf{A}=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$. Show that we have:

a) $\det(\mathbf{A})=\det(\mathbf{A}^T)$.

b) $\det(\mathbf{A}^2)=\det^2(\mathbf{A})$.

3. Solve the equations

a) $\begin{vmatrix}x & -1 & 0 & 0\\ x^2 & 2 & 0 & 0\\ 0 & 0 & 4 & 11\\ 0 & 0 & 1 & 3\end{vmatrix}=0$ **b**) $\begin{vmatrix}1 & 3x & 2 & 2\\ 2 & 4 & 4 & 4\\ 0 & 0 & x & 1\\ 0 & 0 & 1 & 1\end{vmatrix}=0.$

4. Solve the equation

$$\begin{vmatrix}x & 1 & 1\\ 0 & 2 & x\\ x & 3 & 1\end{vmatrix}=-1.$$

5. Without calculating the determinant on the right-hand side of the equality indicated below, Find a matrix $\mathbf{A}\in\mathcal{M}_5$ such that,

$$\det(\mathbf{A})=\begin{vmatrix}1 & 0 & 2 & 1\\ -1 & 1 & 3 & 1\\ 0 & -1 & 4 & 2\\ 0 & 1 & 2 & 3\end{vmatrix}$$

and verifies:

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a) The submatrix formed by the entries of the second row of $\mathbf{A}$ is

$$\mathbf{L}_2 = \begin{bmatrix} 0 & 2 & 0 & 0 & 0 \end{bmatrix}.$$

b) $\mathbf{A}$ is in row echelon form.

c) The matrix $\mathbf{A}$ has no zero entries.

6. Use determinants to find conditions (on the parameters) for the following matrices to be invertible.

$$\mathbf{a}) \quad \begin{bmatrix} a & 1 \\ ab & 1 \end{bmatrix} \qquad \mathbf{b}) \quad \begin{bmatrix} 1 & 1 & -1 \\ 1 & a & 1 \\ 0 & 0 & 1 \end{bmatrix} \qquad \mathbf{c}) \quad \begin{bmatrix} \lambda - 1 & 0 & 0 \\ 1 & \lambda - 2 & 1 \\ 3 & 1 & \lambda - 1 \end{bmatrix}$$

$$\mathbf{d}) \quad \begin{bmatrix} \lambda & -\lambda \\ a & \lambda - b \end{bmatrix} \qquad \mathbf{e}) \quad \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ a & b & c & d & e \end{bmatrix} \qquad \mathbf{f}) \quad \begin{bmatrix} a & ab & 0 \\ ab & a & 0 \\ a & b & a \end{bmatrix}$$

7. Show that the matrix

$$\mathbf{A} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}$$

is invertible iff $a \neq -3b \wedge a \neq b$.

8. Consider the matrices

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} a+b & b & -6c \\ d+e & e & -6f \\ g+h & h & -6i \end{bmatrix}.$$

Knowing that $\det(\mathbf{A}) = -6$, compute $\det(\mathbf{B})$.

9. Use the properties of determinants (alternating function and multilinear function) to prove that:

a) If a matrix has a zero row (column), then its determinant is zero.

b) Se uma matriz tiver duas linhas (colunas) iguais então o seu determinante é zero.

c) If a matrix has two identical rows (columns), then its determinant is zero".

d) If $\mathbf{A} \in \mathcal{M}_n$ Then $\det(\lambda\mathbf{A}) = \lambda^n \det(\mathbf{A})$.

10. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} a & 1 & 1 & f & 0 \\ b & 1 & 2 & g & 1 \\ c & 1 & 1 & h & 1 \\ d & 1 & 2 & i & 0 \\ e & 1 & 1 & j & 0 \end{bmatrix}$$

where, $\det(\mathbf{A}) = 2$.

Compute

a)

$$\begin{vmatrix} a+6 & 1 & 1 & f & 0 \\ b+6 & 1 & 2 & g & 1 \\ c+6 & 1 & 1 & h & 1 \\ d+6 & 1 & 2 & i & 0 \\ e+6 & 1 & 1 & j & 0. \end{vmatrix}$$

b)

$$\begin{vmatrix} a & 3 & 2 & f+1 & 0 \\ b+1 & 6 & 2 & g+2 & 3 \\ c+1 & 3 & 2 & h+1 & 3 \\ d & 6 & 2 & i+2 & 0 \\ e & 3 & 2 & j+1 & 0 \end{vmatrix}$$

c)

$$\begin{vmatrix} a & b & c & d & e \\ 6 & 6 & 6 & 6 & 6 \\ 2 & 4 & 2 & 4 & 2 \\ f & g & h & i & j \\ 0 & 7 & 7 & 0 & 0 \end{vmatrix}$$

11. Let $\mathbf{A} \in \mathcal{M}_3$ such that $\det(\mathbf{A}) = -\frac{1}{3}$. Then,

    a) $\det(-\mathbf{\frac{1}{3}}\mathbf{A}^{-1}) = -\frac{1}{9}$
    b) $\det(-\frac{1}{3}\mathbf{A}^{-2}) = \frac{1}{9}$
    c) $\det(\frac{1}{3}\mathbf{A}^{2}) = -\frac{1}{9}$
    d) None of the above options is correct.

12. Consider the matrix, where α is a real parameter,

$$\mathbf{A}_\alpha = \begin{bmatrix} -1 & 0 & 1 & 0 \\ 4 & 0 & 0 & \alpha \\ 0 & \alpha & -\alpha & \alpha \\ 1 & -1 & \alpha & 4 \end{bmatrix}$$

    Find the values of α for which $\mathbf{A}_\alpha$ is invertible.

13. Let $\mathbf{A}$ e $\mathbf{B}$ square matrices of order 5, such that $|\mathbf{A}| = 3$ and $|\mathbf{B}| = -2$. Compute:
    a) $|\mathbf{A}^T\mathbf{B}|$ b) $|2\mathbf{A}| + |\mathbf{B}^3|$ c) $|-\mathbf{A}\mathbf{B}^2|$ d) $|\frac{1}{2}\mathbf{A}^{-1}\mathbf{B}^T|$

14. Consider the following matricial equation

$$(\mathbf{X}^T\mathbf{A})^T - \frac{1}{2}\mathbf{A}^T\mathbf{X} = \mathbf{I}_n,$$

    where $\mathbf{X}$ is the unknown matrix and $\det(\mathbf{A}) = 2$. if $\mathbf{B}$ is solution of the equation we can conclude:

    a) $\det(\mathbf{B}) = 2^{n-1}$
    b) $\det(\mathbf{B}) = \frac{(-1)^n}{2}$
    c) $\det(\mathbf{B}) = \frac{1}{2^n}$
    d) None of the above options is correct.

15. Calculate the adjoint and the inverse matrices of the following matrices:

a)

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 4 \\ 1 & 2 & 3 \end{bmatrix}$$

b)

$$\mathbf{B} = \begin{bmatrix} 0 & -1/2 & 0 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \\ 1/2 & 0 & -1/2 & 0 \\ 1 & 0 & 1/2 & 1/2 \end{bmatrix}$$

c)

$$\mathbf{C} = \begin{bmatrix} 4 & 3 & 8 \\ 6 & 2 & 5 \\ 1 & 5 & 9 \end{bmatrix}$$

d)

$$\mathbf{D} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad ad - bc \neq 0$$

e)

$$\mathbf{E} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

f)

$$\mathbf{F} = \begin{bmatrix} 4 & 2 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

16. For which values of $a \in \mathbb{R}$ is the matrix $\mathbf{A}$ invertible?

$$\mathbf{A} = \begin{bmatrix} a & 1 & 1 & 2 \\ 0 & a & a & a \\ a & 2a+1 & 2a & 2a+1 \\ 1 & 5 & 3 & 4 \end{bmatrix}.$$

17. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Knowing that $\left(\left(\mathbf{X}^T\mathbf{A}^T\right)^{-1}\right)^T = \mathbf{X}^{-1} - \mathbf{A}^{-1}$, compute, if possible $\det(\mathbf{X})$.

18. Consider the matrices:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b+c & e+f & h+i \\ a+b & d+e & g+h \\ 2c & 2f & 2i \end{bmatrix}$$

Knowing that $\det(\mathbf{A}) = 1$, we can conclude that:

a) $\det(\mathbf{B}) = 2$;

b) $\det(\mathbf{B}) = -2$;

c) $\det(\mathbf{B}) = 0$;

d) None of the above options can be concluded.

19. Consider the matrices $\mathbf{A},\ \mathbf{B} \in \mathcal{M}_3$. If $\det(\mathbf{A}) = 2$ and $\det(\mathbf{B}) = -4$ then we have:

a) $\det(-2\mathbf{A}\mathbf{B}^{-1}) = -8$;

b) $\det(-2\mathbf{A}\mathbf{B}^{-1}) = -4$;

c) $\det(-2\mathbf{A}\mathbf{B}^{-1}) = -\frac{1}{8}$;

d) None of the above options is correct.

20. Consider the matrices $\mathbf{A},\ \mathbf{B} \in \mathcal{M}_n$. Assuming that: $(\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{B})^{-1} = \mathbf{B} - \mathbf{A}$, $\det(\mathbf{A} - \mathbf{B}) = 3$ and that $\mathbf{B}$ commute with $\mathbf{X}$, then we have;

a) $\det(\mathbf{X}) = 3^n$;

b) $\det(\mathbf{X}) = (-3)^n$;

c) $\det(\mathbf{X}) = (-1)^n$;

d) None of the above options is correct.

21. Consider the matrix

$$\mathbf{A} = \begin{bmatrix} 2a & -b & c & -c \\ b & 1 & -1 & b \\ c & 2 & a & 2 \\ 2 & 1 & 2 & d \end{bmatrix}.$$

Find, if possible, a matrix $\mathbf{B} \in \mathcal{M}_6$ that simultaneously satisfies the following conditions:

- $\det(\mathbf{B}) = -\det(\mathbf{A})$
- The fifth row of the matrix $\mathbf{B}$ is [1 2 3 4 5 6]

22. Consider the following matrix $\mathbf{A} = \begin{bmatrix} -2 & -a & 2 & a \\ 3 & -3 & a & 2a \\ a & -5 & 5 & 3a \\ 0 & 0 & 0 & 4a \end{bmatrix}$, where a is a real parameter.

For which values of the parameter a is the matrix $\mathbf{A}$ invertible?

23. Consider the following matrices

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{e} \quad \mathbf{B} = \begin{bmatrix} 6a & 6c & 12b \\ d & f & 2e \\ g & i & 2h \end{bmatrix}$$

Than we can conclude:

a) $\det(\mathbf{B}) = \det(\mathbf{A})$;

b) $\det(\mathbf{B}) = 12\det(\mathbf{A})$;

c) $\det(\mathbf{B}) = 6\det(\mathbf{A})$;

d) None of the above options is correct.

24. Show that

$$\det \begin{bmatrix} 1 & a & a^2 & a^3 \\ a & a^2 & a^3 & 1 \\ a^2 & a^3 & 1 & a \\ a^3 & 1 & a & a^2 \end{bmatrix} = (1 - a^4)^3$$

Solutions

1.

a) -2. **b**) 0. **c**) $-\lambda^3$. **d**) -2. **e**) -2.

f) 4. **g**) 4. **h**) 4. **i**) -2. **j**) -7.

k) 4. **l**) $-2\lambda^2$. **m**) $-2\lambda^3$. **n**) 256. **o**) -1.

p) $\frac{1}{64}$. **q**) -1. **r**) $a(a^2-c^2)+2b^2(c-a)$ **s**) $(z-x)(y-x)(z-y)$

2.

3.

a) $x=0 \vee x=-2$. **b**) $x=\frac{2}{3} \vee x=1$.

4.

$x \pm \frac{\sqrt{2}}{2}$.

5.

for example (there are other equally valid answers):

a)

$$\mathbf{A}=\begin{bmatrix} 1/2 & 0 & 0 & 1 & 1/2 \\ 0 & 2 & 0 & 0 & 0 \\ -1 & 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & 4 & 2 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}$$

b)

$$\mathbf{A}=\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 5 & 2 \\ 0 & 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 & -7 \end{bmatrix}$$

c)

$$\mathbf{A}=\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 3 & 2 \\ 3 & 2 & 1 & -1 & 1 \\ 2 & 2 & 1 & 6 & 4 \\ 1 & 1 & 2 & 3 & 4 \end{bmatrix}$$

6.

a) $a \neq 0 \wedge b \neq 1$. **b**) $a \neq 1$. **c**) $\lambda \neq 1 \wedge \lambda \neq \frac{3+\sqrt{5}}{2} \wedge \lambda \neq \frac{3-\sqrt{5}}{2}$. **d**) $\lambda \neq 0 \wedge \lambda \neq b-a$. **e**) $a \neq 0$. **f**) $a \neq 0 \wedge b \neq 1 \wedge b \neq -1$.

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8.

$\det(\mathbf{B}) = 36$

9.

10.

a) 2. **b**) -36. **c**) 168.

11.

d)

12.

13.

a) -6. **b**) 88. **c**) -12. **d**) $-\frac{1}{48}$.

14.

a)

15.

a)

$$\mathbf{adj}(\mathbf{A}) = \begin{bmatrix} -2 & 2 & -2 \\ 4 & -1 & 0 \\ -2 & 0 & 0 \end{bmatrix}, \quad \mathbf{A^{-1}} = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 1/2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

b)

$$\mathbf{adj}(\mathbf{B}) = \begin{bmatrix} 0 & -1/8 & 0 & 1/8 \\ -1/4 & 3/8 & 1/4 & -1/8 \\ 0 & -1/8 & -1/4 & 1/8 \\ 0 & 3/8 & 1/4 & -1/8 \end{bmatrix}, \quad \mathbf{B^{-1}} = \begin{bmatrix} 0 & -1 & 0 & 1 \\ -2 & 3 & 2 & -1 \\ 0 & -1 & -2 & 1 \\ 0 & 3 & 2 & -1 \end{bmatrix}$$

c)

$$\mathbf{adj}(\mathbf{C}) = \begin{bmatrix} -7 & 13 & -1 \\ -49 & 28 & 28 \\ 28 & -17 & -10 \end{bmatrix}, \quad \mathbf{C^{-1}} = \begin{bmatrix} -1/7 & 13/49 & -1/49 \\ -1 & 4/7 & 4/7 \\ 4/7 & -17/49 & -10/49 \end{bmatrix}$$

d)

$$\mathbf{adj}(\mathbf{D}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \mathbf{D^{-1}} = \begin{bmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$$

e)

$$\mathbf{adj}(\mathbf{E}) = \begin{bmatrix} 5 & -3 & -1 \\ 1 & -1 & 0 \\ -7 & 4 & 1 \end{bmatrix}, \quad \mathbf{E^{-1}} = \begin{bmatrix} -5 & 3 & 1 \\ -1 & 1 & 0 \\ 7 & -4 & -1 \end{bmatrix}$$

f)

$$\mathbf{adj}(\mathbf{F}) = \begin{bmatrix} -2 & -1 & 3 \\ 0 & 3 & -3 \\ 2 & -2 & -6 \end{bmatrix}, \quad \mathbf{F^{-1}} = \begin{bmatrix} 1/3 & 1/6 & -1/2 \\ 0 & -1/2 & 1/2 \\ -1/3 & 1/3 & 1 \end{bmatrix}$$

16.

$a \neq 0$ and $a \neq 1$.

17.

$\det(\mathbf{X}) = \det(\mathbf{A} - \mathbf{I}) = 1$

18.

Option b)

19.

Option d)

20.

Option d)

21.

For example,

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2a & -b & c & -c & 0 \\ 0 & b & 1 & -1 & b & 0 \\ 0 & c & 2 & a & 2 & 0 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 2 & 1 & 2 & d & 0 \end{bmatrix}$$

22.

$$a \in \mathbb{R} \setminus \{0, \pm\sqrt{11}\}.$$

23.

Opção d)